

Tessa Green

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EDUCATION

HARVARD MEDICAL SCHOOL

PHD IN BIOPHYSICS

March 2024 | Boston, MA

Methods and applications of single-cell transcriptomics

MIT

BS IN PHYSICS

February 2016 | Boston, MA

Cum. GPA: 4.9 / 5.0

SUMMARY

Scientific and platform leader in computational biology, specializing in virtual cell evaluation, benchmarking, and model validation. Experienced in leading cross-functional efforts as the sole dry-lab scientist in close partnership with wet-lab teams, executives, and external collaborators. Ensures AI and modeling efforts serve experimental reality and organizational decision-making, while remaining hands-on where it adds the most value.

SKILLS

Machine learning • Data Science •
pytorch • scikit-learn • cluster computing
• single-cell analysis in R (Seurat) and
Python (scanpy) • Python • R

AWARDS

2018	Lynch Fellowship	Harvard
2016	Phi Beta Kappa	MIT
2016	Sigma Pi Sigma	MIT
2011	Thiel Fellowship	(declined)

PATENTS

Green, T. D. "Design for Electricity Transportation and Storage System," US Patent number 8,723,371, May 2014. A modification of flow battery design that would enable simultaneous storage and transport of energy

POSTERS

- scPerturb. NeurIPS Learning Meaningful Representations of Life (LMRL) Workshop 2022
- Prediction of cell-cell communication from scRNA-seq latent spaces. NeurIPS LMRL 2021

EXPERIENCE

KIRAGEN BIO | MACHINE LEARNING SCIENTIST, HEAD OF IMMUNOINFORMATICS

July 2024 - Present | Boston, MA

- Sole dry-lab scientist responsible for end-to-end computational strategy, prioritization and execution
- Led platform-level evaluation and validation of ML-guided gene knockout selection for CAR-T cell therapies targeting solid tumors
- Architected KiraLOGIC, a lab-in-the-loop ML platform developed via strategic partnership with Reflector Bio, integrating virtual cell foundation models with KiraGen-specific fine-tuning, for a 7x improvement in antiproliferative hit ID over baseline.
- Defined experimental benchmarks and performance criteria (e.g. hit-rate thresholds) tying model outputs to wet-lab outcomes and partnership milestones.
- Co-designed high-throughput functional assays with wet-lab teams to maximize biological insight and model improvement
- Authored and communicated the scientific and evaluation narrative for external stakeholders including academic collaborators and investors
- Worked tightly with C-suite on grant writing and strategic planning.

PATIENT-LED RESEARCH COLLABORATIVE | ML CONSULTANT

October 2023 - January 2024 | Remote

- Identified patient subgroups based on binary self-reported symptom data, comparing across machine learning methodologies.
- Led manuscript production and fully executed revisions

HMS DEPARTMENT OF SYSTEMS BIOLOGY | PHD RESEARCHER

October 2017 - October 2023 | Boston, MA

Worked with Professors **Debora Marks** and **Chris Sander**.

- Developed evaluation metrics and benchmarking frameworks for single-cell perturbation and machine learning models.
- Built harmonized database of single-cell perturbation datasets enabling systematic model comparison across conditions (scPerturb).
- Writing lead and project manager for **scPerturb**, a benchmarking resource integrated into the **pertpy** open-source platform.
- Led and coordinated multi-institutional collaborations spanning data generation, modeling, and analysis.
- Applied machine learning methods to immunology and cancer with a focus on careful validation, robustness, and biological plausibility.

HARVARD GRADUATE STUDENT UNION | FINANCE AND BENEFITS COMMITTEE | HGSU-UAW LOCAL 5118

June 2020 - March 2023 | Co-Chair Jan-Dec 2022 | Cambridge, MA

- Co-led policy design and implementation allocating \$1.47M in student benefits
- Streamlined processing and hired/managed 30+ part-time employees

SELECTED PUBLICATIONS

- "scPerturb: Harmonized single-cell perturbation data." S. Peidli*, T.D. Green* et al. Nature Methods 2024. *: co-first authors
- "Pertpy: an end-to-end framework for perturbation analysis." L. Heumos, Y. Ji, L. May, T.D. Green et al, Nature Methods 2025
- "Optimal distance metrics for single-cell RNA-seq populations." Y. Ji, T.D. Green et al. bioRxiv 2023.